Low-Rank Adapters for Efficient Domain Adaptation of Language Models

Chen Wei, Laura Martins

Institute for Machine Learning, Eastlake Research Lab

chen@northbridge.edu, laura@northbridge.edu

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Abstract

Fine-tuning every parameter of a large language model is expensive. We study low-rank adapters that represent the weight update as the product of two small matrices. Across four domain adaptation tasks, rank-8 adapters match full fine-tuning within one point while training 0.4 percent of the parameters and reducing GPU memory by 60 percent.

Keywords: parameter-efficient fine-tuning, LoRA, domain adaptation

1 Introduction

Adapting a pretrained language model to a new domain usually means updating all of its weights. For models with billions of parameters this requires large GPU clusters and produces a full copy of the model for every task.

Parameter-efficient fine-tuning methods train a small number of additional parameters instead. We focus on low-rank adaptation, which freezes the pretrained weights and learns a low-rank update for selected projection matrices.

2 Background

Adapters insert small bottleneck layers between transformer blocks. Prefix tuning learns continuous prompts. Low-rank adaptation instead reparameterizes the update of an existing weight matrix, adding no inference latency after merging.

3 Approach

For a frozen weight matrix W we learn an update B A, where B and A have rank r. Only A and B receive gradients. The update is scaled by alpha over r as shown in Equation 1, which keeps the effective learning rate stable across ranks.

$$h = W_0 x + (\alpha / r) B A x \quad (1)$$

4 Experiments

We adapt a 7B parameter model to biomedical, legal, financial and scientific text and evaluate on the MedQA, CaseHOLD, FiQA and SciQA-2400 datasets. Table 1 compares full fine-tuning with adapters of different ranks.

Table 1: Accuracy of full fine-tuning and low-rank adapters.

Method	Params	MedQA	CaseHOLD
Full fine-tuning	100%	61.3	72.8
LoRA r=4	0.2%	59.8	71.5
LoRA r=8	0.4%	60.6	72.1
LoRA r=16	0.8%	60.9	72.4

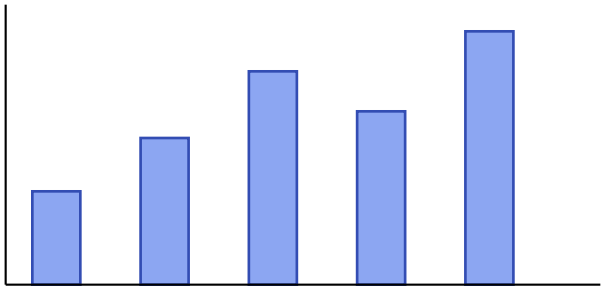

Figure 1: Peak GPU memory during training.

5 Results

Rank-8 adapters are within one point of full fine-tuning on all four datasets. Increasing the rank beyond 16 brings no further improvement. Figure 1 shows that memory use drops by 60 percent.

6 Limitations

We only evaluated models up to seven billion parameters and English-language domains. The optimal rank may differ for larger models or multilingual data.

7 Conclusion

Low-rank adapters are an effective and cheap way to specialize language models. Future work should study automatic rank selection per layer.

References

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